DIGITAL FORENSICS

FINAL INVESTIGATION REPORT

MD5 Hash Recovery

Case Reference No.	*DFIR-2026-004*
Investigator	*Julian Derry*
Date of Analysis	*February 15, 2026*
Prepared For	*Public*
Classification	*Public*

Table of Contents

1. Overview

This report documents the identification and recovery of a provided 32-character hexadecimal hash as part of a password recovery exercise conducted in a Kali Linux environment. The examination involved determining the hash type using automated identification tools and performing a dictionary-based recovery attempt with Hashcat. The objective was to establish the hash algorithm, execute an appropriate recovery method, and document the results in a structured forensic-style report. The analysis resulted in the successful recovery of the plaintext value associated with the target hash.

2. Evidence Handling

Echoed hash to *hash.txt*

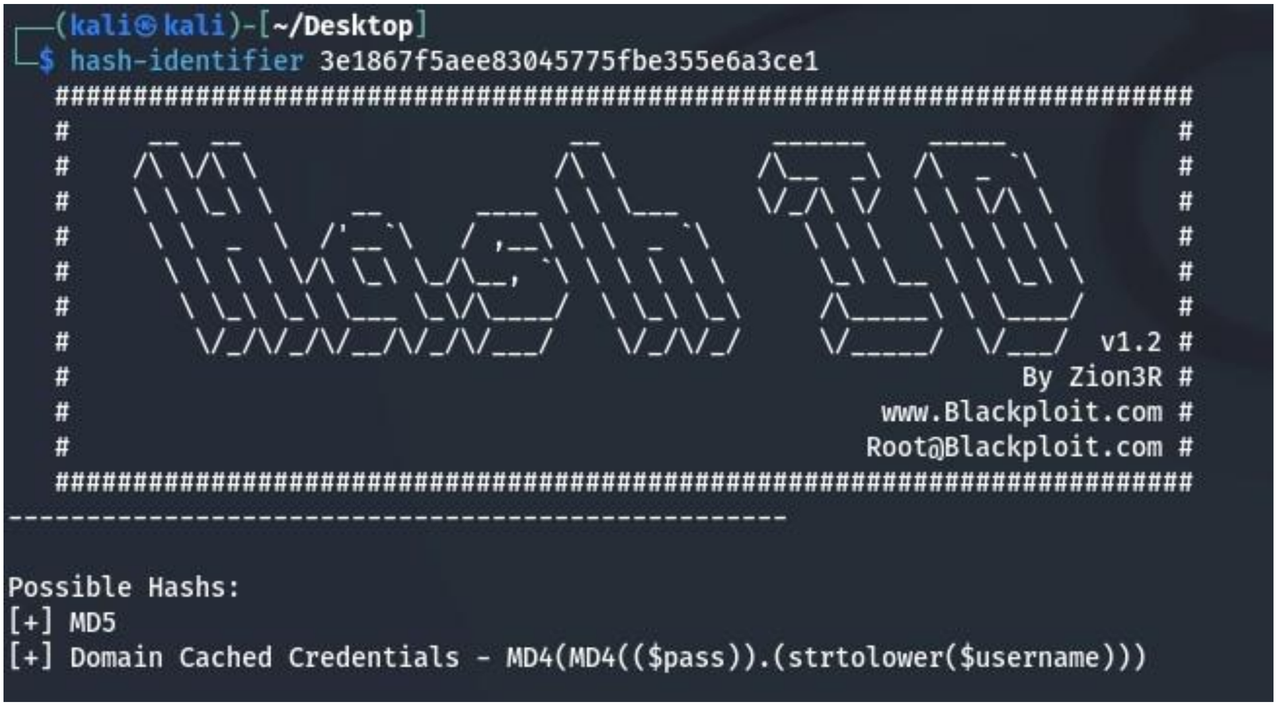
3. Hash Identification

Before launching a recovery attempt, the hash structure was analyzed using two primary tools in the Kali Linux environment:

- **Hashcat Identification:** Running **hashcat --identify** against the string returned 11 potential matches. The top candidates included MD5 (Mode 0) and MD4 (Mode 900).

- **Hash-Identifier:** This tool was used for secondary verification, providing a high-confidence match for MD5.

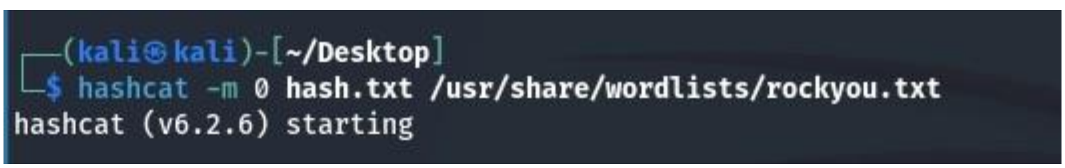
```
┌──(kali㉿kali)-[~/Desktop]
└─$ hashcat --identify 3e1867f5aee83045775fbe355e6a3ce1
The following 11 hash-modes match the structure of your input hash:

      # | Name                                                       | Category
  ======+============================================================+======================================
    900 | MD4                                                        | Raw Hash
      0 | MD5                                                        | Raw Hash
     70 | md5(utf16le($pass))                                        | Raw Hash
   2600 | md5(md5($pass))                                            | Raw Hash salted and/or iterated
   3500 | md5(md5(md5($pass)))                                       | Raw Hash salted and/or iterated
   4400 | md5(sha1($pass))                                           | Raw Hash salted and/or iterated
  20900 | md5(sha1($pass).md5($pass).sha1($pass))                    | Raw Hash salted and/or iterated
   4300 | md5(strtoupper(md5($pass)))                                | Raw Hash salted and/or iterated
   1000 | NTLM                                                       | Operating System
   9900 | Radmin2                                                    | Operating System
   8600 | Lotus Notes/Domino 5                                       | Enterprise Application Software (EAS)
```



4. Recovery Process (Cracking)

Once the hash was identified as MD5, a dictionary attack was performed using Hashcat:

- **Hash Mode:** 0 (MD5)
- **Wordlist:** rockyou.txt
- **Command:** *hashcat -m 0 hash.txt /usr/share/wordlists/rockyou.txt*



The attack was highly efficient, completing in approximately 2 seconds.

```
Dictionary cache hit:
* Filename..: /usr/share/wordlists/rockyou.txt
* Passwords.: 14344385
* Bytes.....: 139921507
* Keyspace..: 14344385

3e1867f5aee83045775fbe355e6a3ce1:beer

Session..........: hashcat
Status...........: Cracked
Hash.Mode........: 0 (MD5)
Hash.Target......: 3e1867f5aee83045775fbe355e6a3ce1
Time.Started.....: Tue Feb 17 07:32:29 2026 (2 secs)
Time.Estimated...: Tue Feb 17 07:32:31 2026 (0 secs)
Kernel.Feature...: Pure Kernel
Guess.Base.......: File (/usr/share/wordlists/rockyou.txt)
Guess.Queue......: 1/1 (100.00%)
Speed.#1.........:   184.8 kH/s (0.79ms) @ Accel:1024 Loops:1 Thr:1 Vec:8
Recovered........: 1/1 (100.00%) Digests (total), 1/1 (100.00%) Digests (new)
Progress.........: 40960/14344385 (0.29%)
Rejected.........: 0/40960 (0.00%)
Restore.Point....: 36864/14344385 (0.26%)
Restore.Sub.#1...: Salt:0 Amplifier:0-1 Iteration:0-1
Candidate.Engine.: Device Generator
Candidates.#1....: holabebe -> loserface1
Hardware.Mon.#1..: Util: 19%

Started: Tue Feb 17 07:32:10 2026
Stopped: Tue Feb 17 07:32:32 2026
```

5. Results

The hash was successfully recovered, and the plaintext value was determined to be **"beer".**

Target Hash: 3e1867f5aee83045775fbe355e6a3ce1

Recovered Plaintext: beer

5. Tools Used

- **Hashcat**
- **Hash-Identifier**
- **Kali Linux**
- **rockyou.txt**

6. Limitations

No limitations were documented in the original report. I ran:

7. Conclusion

The hash was successfully identified as MD5 and recovered through a dictionary-based attack using Hashcat. The plaintext value obtained was **"beer".**

8. Appendix

Target Hash: 3e1867f5aee83045775fbe355e6a3ce1

Recovered Plaintext: beer

Command Used: hashcat -m 0 hash.txt /usr/share/wordlists/rockyou.txt